

SEQUENCE Function in Excel

1. What is SEQUENCE?

The SEQUENCE function creates a list or grid of numbers automatically from a single formula. It solves the problem of typing or dragging numbers by hand, such as 1, 2, 3 or 1001, 1002, 1003. A Data Analyst uses it to create serial numbers, ID lists, and index numbers that update when the data changes.

2. Syntax

`=SEQUENCE(rows, [columns], [start], [step])`

Argument	Required?	What it accepts
rows	Required	How many rows of numbers to create (a whole number of 1 or more)
columns	Optional	How many columns to create. The default is 1.
start	Optional	The first number in the sequence. The default is 1.
step	Optional	The amount added each time. The default is 1. A negative step counts down.

Excel fills the grid row by row. `=SEQUENCE(2,3)` gives 1, 2, 3 in the first row and 4, 5, 6 in the second row. `=SEQUENCE(5,1,10,5)` gives 10, 15, 20, 25, 30 in one column.

3. Simple Example

Generate the numbers 1 to 5 in a single column:

	A
1	Result
2	1
3	2
4	3
5	4
6	5

Formula in A2 only: `=SEQUENCE(5)`

Excel creates 5 numbers, starting at 1 and adding 1 each time. It places them in A2 to A6 automatically. This is called "spilling".

Result: 1, 2, 3, 4, 5

4. Another Example

Create order numbers that start at 1001 for four customers:

	A	B
1	Customer	Order ID
2	Aisha	1001
3	Rahim	1002
4	Nabila	1003
5	Tanvir	1004

Formula in B2 only: `=SEQUENCE(4, 1, 1001)`

Excel creates 4 rows and 1 column, starting at 1001 and adding 1 each time. The step is left out, so it uses the default of 1. **Result: 1001 to 1004**

5. When to Use It

- Create serial numbers or ID numbers for a list of records
- Create month numbers 1 to 12 for a yearly report
- Build a list of dates by using a date as the start value (format the cells as dates)
- Create index or helper numbers for other formulas
- Generate quick sample numbers for practice or testing

6. Important Things to Remember

- SEQUENCE is a dynamic array function. One formula in one cell fills many cells. This is called "spilling".
- Numbers fill row by row: left to right across the first row, then down to the next row.
- The cells where the numbers will spill must be empty. If they are not, Excel shows `#SPILL!`.
- If you leave out `columns`, `start`, or `step`, Excel uses 1 for each one.
- `rows` and `columns` must be 1 or more. The `step` can be negative to count down, for example `=SEQUENCE(5,1,10,-2)` gives 10, 8, 6, 4, 2.
- SEQUENCE is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Getting `#SPILL!` because cells in the spill area already contain data. **Fix:** Clear those cells, or move the formula to an empty area.

Mistake: Trying to edit one of the spilled numbers. **Fix:** Only the first cell holds the formula. Edit that cell, and the whole result updates.

Mistake: Mixing up rows and columns, such as `=SEQUENCE(1,5)` (one row) when a column was needed. **Fix:** Remember the order: rows first, then columns.

Mistake: Trying to set a step without a start, such as `=SEQUENCE(5,1,2)`, which starts at 2 with a step of 1. **Fix:** Fill the arguments in order: `=SEQUENCE(5,1,1,2)`.

8. Data Analyst Tip

Use SEQUENCE for numbering that adjusts itself. For example, `=SEQUENCE(COUNTA(A2:A100))` creates one number for each filled cell in A2:A100, so the numbering grows or shrinks with your data. This is much safer than typing numbers by hand or dragging them down.